

Additive Manufacturing of a Lightweight Rotor for a Permanent Magnet Synchronous Machine

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Abstract—Additive Manufacturing (AM), also known as 3D printing, is a relatively new technology which enables the toolless production of components and entire assemblies directly from a CAD file. Today, the technology is still not widely used in industrial production. It is mainly limited to special applications, although it shows great potential. In this paper, first approaches are shown to apply AM to the production of rotors for permanent magnet synchronous machines (PMSM). The possibilities of a lightweight design with a low moment of inertia as well as the influence on the magnetic anisotropy for an improved sensorless control of PMSM are pointed out. The results clearly demonstrate the great potential of additive manufacturing in electrical engineering applications.

Keywords—additive manufacturing, 3D printing, PMSM, material properties, sensorless control

I. INTRODUCTION

Additive manufacturing processes create components and assemblies in layers and without the use of forming tools [1]–[3]. In this way, very complex component geometries can be designed and produced which show, for example, undercuts, integrated lattice structures, bionic structures, or free-form surfaces, which are not economically feasible with conventional production processes [1],[4]. Due to their advantages, additive manufacturing processes gain an increasingly growing interest in industry and research [5]. So additive manufacturing processes promote the implementation of the Industry 4.0 strategy, secure jobs, shorten transport routes and provide opportunities for new business models. A successful application of additive manufacturing processes contains the consideration of their design rules [6]. Some of such rules are already known for additive manufacturing processes [7]. These rules give instructions for individual design aspects, such as for example the thickness of a wall. The rules are usually process-specific (e.g. for selective laser beam melting) and address only a few design aspects.

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Therefore, design rules for additive manufacturing processes were developed which allow a comprehensive usage and transferability to individual components in electrical engineering like PMSM rotors [3]. Moreover, additive manufacturing processes show a great potential for function integration into components like a sensorless control. In this paper, investigations by measurements of a prototype with an additively manufactured rotor shaft demonstrate the feasibility of the technology in electrical machines. For lack of specialised ferromagnetic materials for AM, H13 was identified as the most suitable conventional AM material. Besides, specimens were developed and produced with AM to investigate the resulting mechanical and electromagnetic properties to ensure a scientifically sound approach.

II. MATERIAL ANALYSIS

A. SLM Process

The specimens and the rotor shaft were built with the selective laser melting (SLM) process. This process allows for time-efficient, flexible and toolless manufacturing of complex metal components, based on the principle of a precise melting, cooling and coating of metal powder. Through metallurgically bonded connections between layers, a solid 3D-shaped component can be realized. This component can be manufactured directly from CAD data with the help of a computer-controlled laser.

B. Mechanical Analysis

The tensile strength and elongation at rupture of the solid material H13 (1.2344) were determined by tensile tests on R2 specimens, which were built parallel to the load direction. The geometry of the R2 specimens is based on the DIN EN ISO 6892-1 standard. The adjusted geometry of the specimens saves powder material and machine running time.

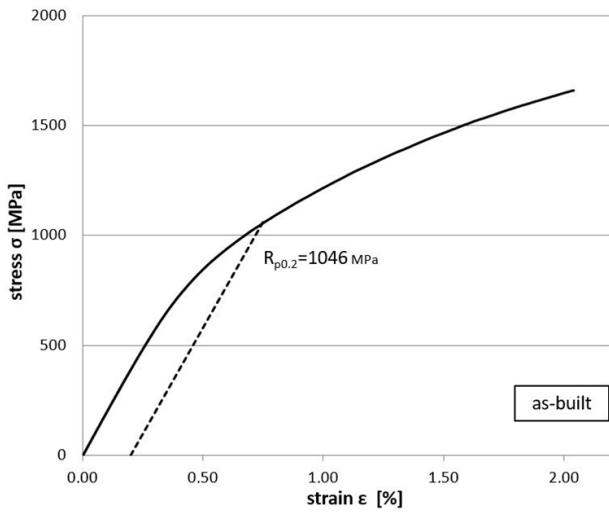


Figure 1: Results of tensile tests on H13 specimens

The significance of the R2 samples as well as the reproducibility of mechanical properties could be verified already in various projects [8],[9]. The specimens showed an 8 mm long and 3 mm wide measuring range of 1.5 mm thickness. The tensile tests were carried out at room temperature on a servo-hydraulic MTS tensile test system at a constant speed of 0.025 mm/s. The elongations were recorded with an extensometer at a measuring length of 3 mm. The average tensile strength of five tests was 1650 MPa with an elongation at rupture of 2%. The results are shown in Figure 1. The investigated specimens were tested without a heat treatment directly after the additive manufacturing process in the so-called "as-built" state. The Vickers hardness varies depending on the distance of the building platform between 670 HV5 and 620 HV5. Higher layers in the building space show a higher hardness than lower layers. This results from heat dissipation through the lower layers which affects the heat treatment of the lower layers [8].

C. Electromagnetic Analysis

For the investigation of the electromagnetic properties, ring cores were produced with an outer diameter of 62 mm and a cross-section of 6 x 6 mm. The cross-section is composed of a solid frame and a structured interior. Five different variants were produced. One ring was made entirely from solid material, one hollow variant with unmelted powder inside the solid frame and three variants with different lattice structure base cells inside the solid frame. The lattice structures are shown in Figure 2. The base cells have a size of 2 x 2 x 2 mm and a bar diameter of 0.2 mm. They were manufactured to be positioned flat in the building chamber to minimize the construction time. To identify the magnetic properties of the AM material H13, the ring cores were wound with a primary excitation winding with 200 turns and a secondary measurement winding with 100 turns to apply a magnetic field and measure the average magnetic flux density. The magnetic measurements were carried out on a Brockhaus MPG200 device. The characterization of the ring cores revealed a hard magnetic behavior. Figure 3 shows the results of the hysteresis measurement.

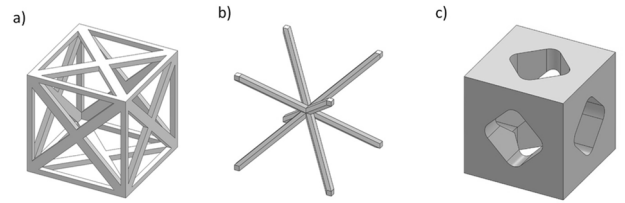


Figure 2: Different investigated lattice structures a) face-centered cubic base (fcc) cell b) body-centered cubic (bcc) base cell c) pyramid base cell

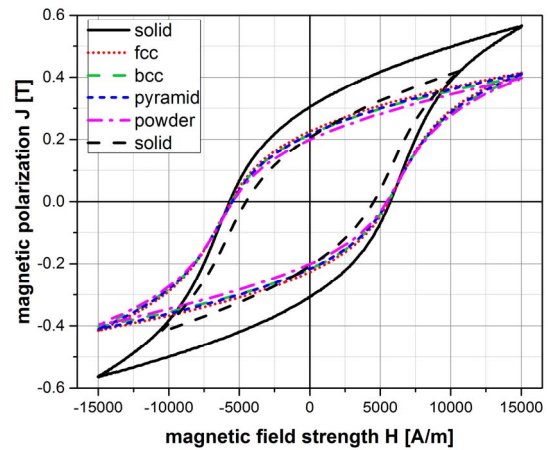


Figure 3: Graphs of the hysteresis of the not heat-treated solid and structured ring cores

Comparing the hysteresis curves, no significant difference between the structured rings can be recognized. Compared to the solid ring, the structured rings show a smaller residual magnetism J_R after magnetization to the same magnetic field strength H (black solid line) and a higher coercive field strength H_{CJ} after magnetization to the same polarization J . The poor magnetic properties can be ascribed to a martensitic microstructure with parts of austenite. Heat treatment was performed to improve the magnetic properties with a ferrite-pearlite microstructure. The ring cores were heat-treated in a tube furnace under argon atmosphere to prevent oxidation. The temperature was 780 °C for 20 min with a heating rate of 300 °C/h and a slow cooling down in the furnace. The hysteresis curves of the heat-treated ring cores are shown in Figure 4. The solid ring has a higher saturation magnetic polarization than the structured rings. A difference between the hysteresis of the fcc and bcc ring is not visible. From measurements of the magnetization curves, the relative permeability was calculated. The maximum differential relative permeability $\mu_{r, \max} = \max(dB / d(\mu_0 \cdot H))$ for the not heat-treated and heat-treated ring cores is listed in Table 1. The heat treatment resulted in a significant improvement of the permeability. Measurements of the total losses P_s were carried out at two different frequencies: 50 Hz (Figure 5) and 1000 Hz (Figure 6). At 50 Hz, the structured rings have higher losses than the solid ring. With increasing frequency, the eddy current losses become the dominant contribution to the total losses.

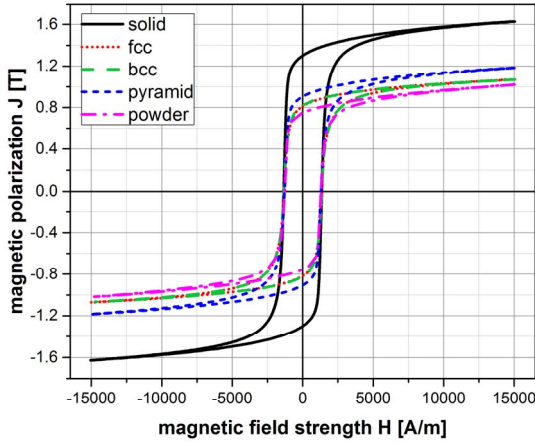


Figure 4: Graphs of the hysteresis of the heat-treated solid and structured ring cores

Table 1: Maximum permeability of the not heat-treated (NHT) and heat-treated (HT) ring cores

	solid	fcc	bcc	pyramid	powder
$\mu_{r, \max}$ (NHT)	32.71	24.78	24.10	24.24	23.40
$\mu_{r, \max}$ (HT)	484.98	288.29	292.10	344.71	298.45

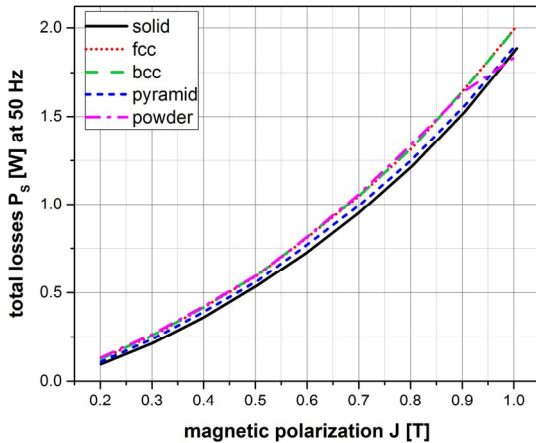


Figure 5: Total losses of heat-treated ring cores at 50 Hz

At 1000 Hz, the total losses of the solid ring are higher than the total losses of the structured rings, because the eddy current losses can spread in a more undisturbed way. The resistance of the ring cores was determined with the four-point probe resistivity measurement method. The current was applied with two screw contacts, and the voltage was measured between two alligator clips. With the measured resistance R , the cross-section A and the length between the voltage measuring points l , the area-related resistivity is calculated $\rho = (R \cdot A / l)$. The results of the calculation are listed in Table 2. The solid ring has the lowest and the powder-filled ring the highest resistivity.

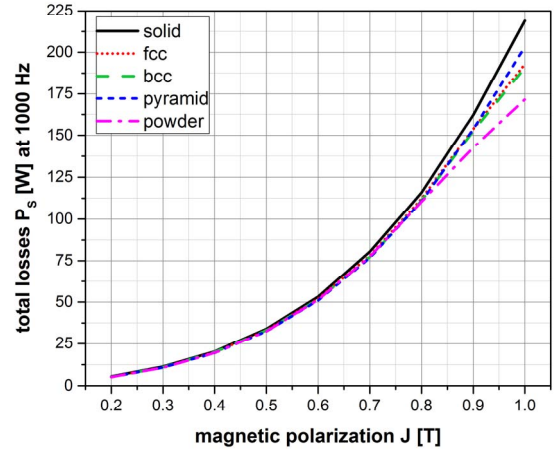


Figure 6: Total losses of heat-treated ring cores at 1000 Hz

Table 2: Area-related resistivity of the heat-treated ring cores

	solid	fcc	bcc	pyramid	powder
ρ [$\Omega \cdot \text{mm}^2/\text{m}$]	0.25	0.43	0.41	0.35	0.45

III. REDESIGN OF ROTOR SHAFT

A. Suitable Design for Production

The design of the rotor was optimized in terms of a production-ready lightweight design. For this purpose, the orientation of the rotor in the building space was adjusted. Its axis of rotation correlates with the z-axis of the SLM system. This results in a higher precision of the rotor, because the step effect does not occur on the cylindrical shell surface. The step effect normally occurs when manufacturing curved components in layers. If a curved surface is oriented perpendicularly to the z-axis of the machine, it is approximated by rectangles. This approximation becomes noticeable by a higher surface roughness of the components or deviations. Besides, slopes from 45° instead of overhangs were introduced. These 45° slopes can be produced without supports, thus reducing the material demand. At the same time, the reprocessing effort is reduced, because the manual removal of the support is decreased to a minimum. In order to reduce the weight while maintaining strength, cavities and lattice structures were implemented into the rotor. This lowers the moment of inertia of the rotor about 23%. In addition, gaps and holes were provided to remove the unplasticized metal powder in the area of lattice structures and cavities. Hereby, the powder can be reused and costs can be saved. The holes for the powder removal were designed in a teardrop-shaped geometry to avoid supports during the manufacturing process. The entire rotor design was checked for possible crack growth starting points. The potential starting points were defused by introducing radii on sharp edges of the rotor geometry.

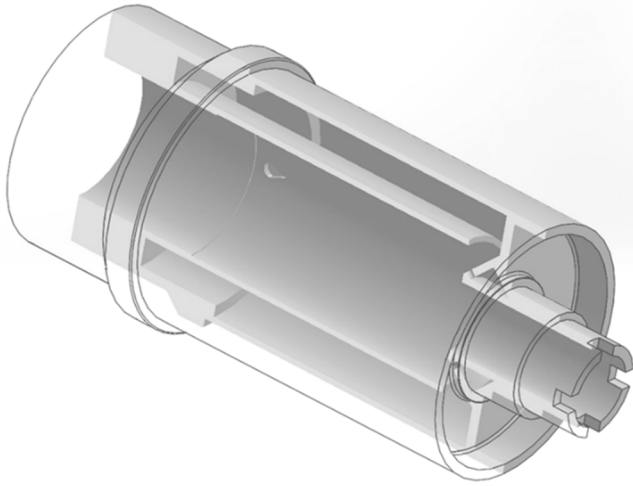


Figure 7: Additively manufactured lightweight rotor shaft

In addition, the scheduled threads were removed from the original design, because the cutting of threads is technically more reproducible and cost-effective in a conventional post-processing step. A lateral cut of the final lightweight rotor is shown in Figure 7. The visible cavities are filled with lattice structures to ensure a high rigidity of the rotor and the most favorable lightweight properties.

B. Sensorless Control

Controlling PMSM requires an accurate detection of the rotor position. The rotor position measurement by means of mechanical encoders increases the construction volume and cost of the electrical machine, whereas sensorless techniques estimate the rotor position based on the terminal voltages and currents, so that no additional components are required. Since the PMSM is used as a sensor itself, this method is also called sensorless control [10]. But not every PMSM is suitable for rotor position estimation methods in the whole speed range. At low speed and at standstill, the electromagnetic characteristics of the machine must meet special requirements that go beyond the usual requirements for performance, losses and acoustic noise. Here, the rotor position estimation is based on a high frequency signal injection, which is only applicable, if a sufficient magnetic differential anisotropy of the machine ($L'_d \neq L'_q$) exists in every operating point. The torque dependency of the anisotropy is a crucial design criterion. It can be influenced by the electromagnetic design, especially by the design of the rotor core.

AM opens new possibilities and additional degrees of freedom for the rotor design. One example is the realization of flux barriers; instead of removing all material in these sections, lattice structures can be used to maintain the mechanical strength of the rotor structure. Therefore, more flux barriers or more complex structures can be used to improve the sensorless control capabilities.

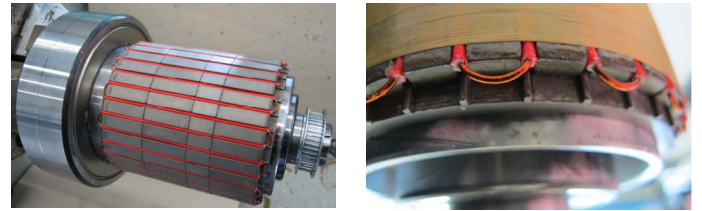


Figure 8: Prototype with short-circuited coil in a meandering pattern in the rotor. The conductors are fixed between the permanent magnets. Rotor coil distribution without bandage (left). Winding overhang of the final rotor with a bandage (right)

These advantages can only be utilized, if the permeability of the material is sufficiently high. But this requirement is not met by the H13 material in the presented prototype. For this reason, a different approach was used. Instead of changing the design of the rotor yoke, an additional copper damping coil in a meandering pattern is arranged around the magnets with one conductor in each pole gap, as shown in Figure 8. A bandage is used to protect the magnets and the coil against radial forces. The rotor coil reduces the effective high frequency inductance in the magnetic direct axis L'_d , while the inductance in the quadrature axis L'_q is not changed. Therefore, the differential magnetic anisotropy is increased which improves the sensorless rotor position estimation. The concept is based on [11] and [12].

IV. DETERMINATION OF MOTOR CHARACTERISTICS

In the following, the prototype with an AM shaft and a short-circuited coil in the rotor is compared to the conventionally manufactured reference machine. The measured values of the rated torque and the voltage constant are almost identical, as listed in Table 3. Since the rotor yoke in both, the reference machine and the prototype, are made of electrical steel, this behavior is expected. The minor differences are assumed to be caused by fluctuations of permanent magnets' material properties and by various manufacturing effects. The lightweight design increases the angular acceleration by 32.2%. In summary, it can be stated that the AM design significantly improved the mechanical dynamics of the PMSM without drawbacks in other performance parameters such as maximum torque or efficiency.

Furthermore, the magnetic anisotropy could be increased from basically zero to almost 20% at 500 Hz. The measured dependency on the frequency is shown in Figure 9. This allows an accurate rotor position estimation at low speeds. The results in Figure 10 prove that the resulting estimation error of the electrical angle stays below 4° even at high load currents. Therefore, this concept is very beneficial for sensorless control applications especially with surface-mounted magnet configurations. AM offers various possibilities to embed such a rotor coil in a more suitable way, such as a positive-fit connection between coils and rotor, so that no bandage is required for radial force protection.

Table 3: Comparison between the reference machine and the prototype

	reference	prototype	
angular acceleration (in rad/s ²)	3871	5118	+32.2 %
torque at I ≈ 70 A (in Nm)	69.74	71.98	+3.2 %
voltage constant (in Vs)	0.891	0.910	+ 2.1 %

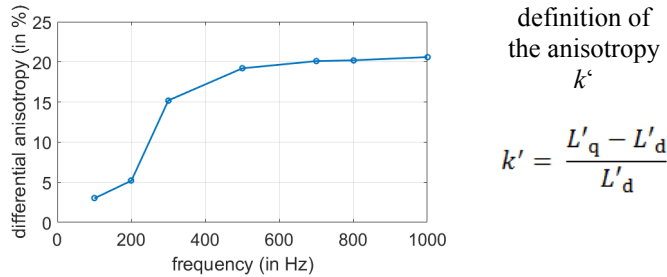


Figure 9: Measured differential anisotropy as a function of the hf frequency

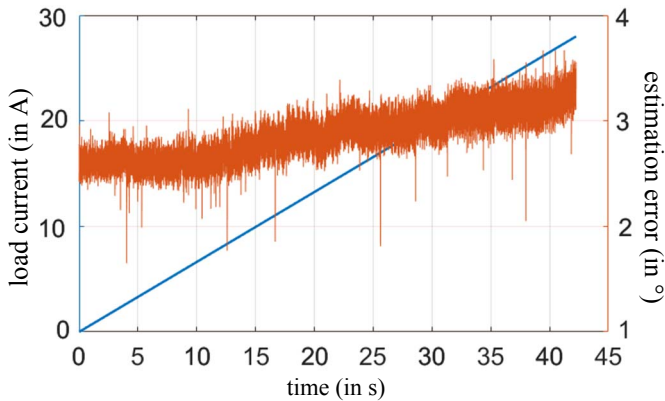


Figure 10: Rotor position estimation error (red) at different levels of load current (blue)

V. CONCLUSION

The results show that the magnetic properties of H13 can be significantly improved by a heat treatment, so that its permeability nearly reaches the level of SMC materials. Finally, an improved new rotor with a lightweight design was developed and manufactured for a prototype machine. Compared to the conventional design, the mass of the rotor is reduced by 25% and the moment of inertia decreases by 23%. This leads to a shorter acceleration time by 23.2%. Additionally, sensorless control was made possible for this machine by means of rotor coils. Future materials for AM with better magnetic properties will unlock the full potential of the additional degrees of freedom in the rotor design. The results clearly demonstrate the great potential of additive manufacturing in electrical engineering applications. For future studies, more investigations of the electromagnetic behavior of additively manufactured materials are needed.

Besides, realizable function integration like cooling effects or lightweight designs are required to establish additive manufacturing as a standard process in industrial process chains.

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References

- [1] P.C. Joshi, R.R. Dehoff, C.E. Duty, W.H. Peter, "Direct Digital Additive Manufacturing Technologies Path Towards Hybrid Integration", Future of Instrumentation International Workshop (FIIW) 8-9.10.2012, p 1-4, Gatlinburg, IEEE, DOI 10.1109/FIIW.2012.6378353, 2012
- [2] G.N. Levy, R. Schindel, J.P. Kruth.: "Rapid Manufacturing and Rapid Tooling with layer manufacturing technologies – State of the art and future perspectives." CIRP Annals - Manufacturing Technology, Vol. 52 No. 2, 589 – 609, 2003
- [3] D. Zimmers, G.A.O. Adam, "Design rules for additive manufacturing" Konstruktionsregeln für Additive Fertigungsverfahren. Zeitschrift Konstruktion, Ausgabe 07/08, Springer Verlag, S. 77-82
- [4] R. Hague, I. Campbell, P. Dickens, "Implications on design of rapid manufacturing." Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science 2003, 217:25, 2003
- [5] S. Leuders, M. Thöne, A. Riemer, T. Niendorf, T. Tröster, H.A. Richard, H.J. Maier, "On the behavior of titanium alloy TiAl6V4 manufactured by selective laser melting: Fatigue resistance and crack growth performance." International Journal of Fatigue, Volume 48, Elsevier, 2013
- [6] B. Vayre, F. Vignat, F. Villeneuve, "Designing for Additive Manufacturing." 45th CIRP conference on Manufacturing Systems 2012, 16-18.05.2012, Athen, Elsevier, 2012
- [7] G. Adam, D. Zimmer "On design for additive manufacturing: evaluating geometrical limitations" Rapid Prototyping Journal, Vol. 21, Iss. 6, 2015, p. 662-670
- [8] M.J. Holzweissig, A. Taube, F. Brenne, M. Schaper, T. Niendorf, "Microstructural Characterization and Mechanical Performance of Hot Work Tool Steel Processed by Selective Laser Melting" Metallurgical and Materials Transactions B; DOI: 10.1007/s11663-014-0267-9; published Januar 2015
- [9] S. Leuders, T. Lienenke, S. Lammers, T. Tröster, T. Niendorf, "On the fatigue properties of metals manufactured by selective laser melting – The role of ductility" Journal of Materials Research, DOI: <http://dx.doi.org/10.1557/jmr.2014.157>, 2014
- [10] K. Wiedmann, A. Mertens, "Self-sensing control of PM synchronous machines including online system identification based on a novel MRAS approach," in 2012 IEEE Symposium on Sensorless Control for Electrical Drives (SLED), 2012, pp. 1–8.
- [11] F. Quattrone, B. Ponick, "Active Differential Inductance Control of Permanent Magnet Synchronous Machines Using Short-Circuited Rotor Coils," in Vehicle Power and Propulsion Conference (VPPC), 2015 IEEE, pp.1-7, 19-22 Oct. 2015.
- [12] K. Wiedmann and A. Mertens, "Novel MRAS approach for online identification of key parameters for self-sensing control of PM synchronous machines," Power Electronics and Motion Control Conference (EPE/PEMC), 2012 15th International, Novi Sad, 2012, pp. LS4b-1.2-1-LS4b-1.2-8.